

Energy Savings of a Four-Drive Electric Bus With Two Cascaded Smart Integrated Drives on Each Axle

Kaibo Li^{ID}, Alain Bouscayrol^{ID}, *Member, IEEE*, Walter Lhomme^{ID}, Betty Lemaire-Semail^{ID}, *Member, IEEE*, Kurt Stockman^{ID}, and Charles Brocart

Abstract—Multidrive systems are explored as a replacement for single-drive setups in automotive applications. By allowing the selection of one or several drives based on the vehicle operation, a reduction in energy consumption is expected. This article proposes a four-drive traction topology for an urban electric bus, featuring two smart integrated drives in cascade on both driven axles. A single-drive-based electric bus with a validated model serves as a reference. The four-drive system is then incorporated into this e-bus. Energetic macroscopic representation (EMR) is employed to organize the model and control both bus topologies. An optimal energy management strategy is devised to minimize energy consumption. The four-drive electric bus achieves a reduction in energy consumption of up to 32% for various urban bus driving cycles. This energy saving is attributed to increased braking energy recovery and an optimal distribution of electric drive torques. A sensitivity analysis on mass variation indicates that the proposed system remains efficient for e-bus applications, even with an increase in mass and volume.

Index Terms—Electric machines, electric vehicle, energetic macroscopic representation (EMR), energy management, road vehicle control.

ABBREVIATIONS

CE2I	Convertisseur d'Énergie Intégré Intelligent (smart integrated drive).
EM	Electric machine.
EMR	Energetic macroscopic representation.
MCM	Modular cascade machines.
SoC	State of charge.
SORT	Standardized on-road tests.

Manuscript received 17 April 2024; revised 13 June 2024 and 1 August 2024; accepted 29 August 2024. Date of publication 4 September 2024; date of current version 3 February 2025. This work was supported by the Intelligent Integrated Energy Converter (CE2I) and Electrical Energy 4.0 (EE40) Projects. CE2I and EE40 are Co-Financed by European Union with the financial support of the European Regional Development Fund (ERDF), French State, and French Region of Hauts-de-France. It is also co-supported by the Récupération d'Énergie de Métro pour une Université Soutenable (REMUS) and Scalability of powerTrain for electrified Vehicles of an Eco-campus (STeVE) Projects of the Campus of University with Mobility based on Innovation and Carbon Neutral (CUMIN) Program of the University of Lille, France. (*Corresponding author: Alain Bouscayrol.*)

Kaibo Li is with the L2EP, University of Lille, 59655 Villeneuve d'Ascq, France, and also with the University of Warwick, CV4 7AL Coventry, U.K. (e-mail: likaibo_hit@163.com).

Alain Bouscayrol, Walter Lhomme, and Betty Lemaire-Semail are with the L2EP, University of Lille, 59655 Villeneuve d'Ascq, France (e-mail: alain.bouscayrol@univ-lille.fr; walter.lhomme@univ-lille1.fr; betty.semail@univ-lille.fr).

Kurt Stockman is with the Department of Electromechanical, Systems and Metal Engineering Ghent University, 9052 Ghent, Belgium (e-mail: kurt.stockman@ugent.be).

Charles Brocart is with Metropole Européenne de Lille (MEL), 59040 Lille, France (e-mail: cbrocart@lillemetropole.fr).

Digital Object Identifier 10.1109/TTE.2024.3454443

NOMENCLATURE

E	Energy.
F	Force.
M	Mass.
P	Power.
T	Torque.
Ω	Rotational speed.
c	Controller.
g	Gravity coefficient.
i	Current.
k	Gear ratio.
f	Frictional coefficient.
r	Ratio of braking energy to traction energy.
t	Time.
u	Voltage.
v	Velocity.
α	Slope of the road.
γ	Power flow coefficient of electric drive.
η	Efficiency.
κ	Discrete subscribe.
λ	Power flow coefficient of the vehicle.

SUBSCRIPTS

aux	Auxiliary devices.
bat	Battery.
bk	Brake.
ed	Electric drive.
elec	Electrical component.
env	Environment.
ev	Electric vehicle.
f	Front axle.
i	$i = 1$ or 3 , the number of CE2I drive.
k	$k = [f \text{ or } r]$, representing the front or rear axle.
max	Maximal value.
mea	Measurement.
mech	Mechanical component.
min	Maximal value.
N	Rated value.
ocv	Open-circuit voltage.
pt	Powertrain.
trac	Traction.
trans	Transmission.
r	Rear axle.
ref	Reference signal.
reg	Regenerative component.

- res Resistive component.
wh Wheel.
s $s = 1$ or 2 or 3 or 4 , CE2I drive number.
sys System component.

I. INTRODUCTION

ROAD transport is the fastest growing source of carbon emissions. This trend emphasizes the urgent need for sustainable solutions [1]. Although buses are fewer in number compared to cars, their impact on emissions is significantly higher [2]. Therefore, bus electrification is crucial for sustainability. In 2021, zero-emission buses had a sales share exceeding 10% in various regions. China led the way with a substantial share of 34%. U.K. accounted for 11%, while the EU saw slightly more than 10% of the sales share [3].

E-buses are driven by an electric machine (EM), which is expected to possess high power density, high efficiency, low cost, good overload capability, and high reliability [4]. In recent decades, multimachine systems [5], [6], [7], multi-phase machines [8], [9], [10], and special structure machine systems [11], [12], [13] have been proposed to enhance the performance of conventional three-phase induction machines and permanent magnet synchronous machine drives. The main idea is to introduce additional degrees of freedom for better and more reliable performance. For multimachine systems, the configuration typically takes distributed or centralized forms, determined by the number of electrical and mechanical ports.

Centralized systems have one mechanical port to the transmission, such as a classical single drive. They require fewer structural modifications when replacing a single-drive system. Various connection configurations are possible for centralized systems. In [14], [15], and [16], several modularly designed EMs are cascaded by sharing a common shaft. This means that the speeds of the EMs are coupled, but their torques are independent. This setup can replace a single-drive system while maintaining the same electrical and mechanical ports. It has been shown to save 5%–12% of energy with an optimal strategy. However, this increases the volume, mass, and cost of the electric drive. In some studies, centralized systems use a set of gear pairs. The EM shafts are linked to a geartrain, with the cumulative output power transmitted through a common mechanical shaft [17], [18], [19]. Different gear ratios can be chosen based on driving conditions. EMs can operate within different speed ranges, improving efficiency. However, a geartrain adds mass, volume, and mechanical losses. Another option is to use a planetary gearbox. In [20] and [21], two EMs are connected to the sun and the ring of a planetary gearbox, with the carrier as the unique mechanical port. This topology decouples speeds (speed-addition) while coupling torques. In [22], a dual-machine system is coupled by a planetary geartrain and a synchronizer, allowing shifts between torque-addition and speed-addition operations. This enhances operational modes but adds considerable complexity.

Distributed systems, such as wheel-side and in-wheel setups, offer multiple output mechanical ports. For example, placing an electric drive on both front and rear axles improves driving abilities [23] and enhances braking energy recovery [24]. Commercial models with distributed systems include Tesla

Model 3, Audi Q4 E-tron 50, Volvo XC40, and Mercedes-Benz (580). In-wheel or wheel-side electric drives, such as those in BYD K9 and Ford eWheelDrive, integrate EMs directly into wheels, without a differential and saving space [25]. However, this integration complicates EM design, limiting power and torque by wheel size [26] and requiring high vibration control [27]. Alternatively, wheel-side EMs with a gearbox inserted between the wheel and high-speed EM can be used [28].

To sum up, the benefits of a centralized system come from the power split optimization of the EMs. One or several EMs can be selected based on the operation of the vehicle. Moreover, the operating points of the EMs can be optimized, which improves the overall efficiency. Except for the gain from power split, a distributed system also takes benefits from the energy recovery ability of the additional driven axle. In addition, a multimachine system exhibits higher reliability compared to a single-drive system. To the authors' best knowledge, the combination of a centralized system and a distributed system has not yet been studied.

The objective of this article is to propose a four-drive system for an urban electric bus, where two smart integrated drives (CE2I, French acronym) are in cascade on both driven axles. While this new topology could potentially lead to energy savings, quantifying the energy gain is challenging. At the system level, the differences are influenced by various factors such as the characteristics of the vehicle under study, design constraints of the components, control laws, and energy management strategies. Therefore, a specific method has been developed to compare different vehicle topologies and quantify the potential energy savings fairly. This analysis could be crucial in deciding whether to invest in such a new vehicle topology. The studied CE2I drive integrates a 2×5 phase permanent magnet synchronous machine and a GaN-based inverter into a module with a dc electrical port [29]. The modular design makes the CE2I drive have the plug-and-play property. Several CE2I drives can be directly connected to the same dc bus. The house, shaft, and cover of the EM can be modularly designed with specific interfaces so that it is convenient to form a multidrive system. The CE2I drive can be adapted to various power levels, and different power demands can be met by the drive number.

The contribution of this work is summarized as follows.

- 1) A composite multidrive configuration is proposed for an e-bus, merging a centralized system with a distributed one. In this new topology, a cascaded dual-machine system is placed on both driven axles.
- 2) A dedicated flexible control organization based on EMR is developed to take benefits of this new multidrive system, which enables the switching between different topologies. An original optimal online energy management is proposed to maximize the benefits of the proposed topology.
- 3) The energy-saving potential of the proposed e-bus topology is quantified through a comparison with a reference e-bus, and up to 32% of energy can be saved.
- 4) A dedicated method was proposed to ensure a fair comparison with the reference e-bus. This method respects the same constraints, ensuring that any gains come from

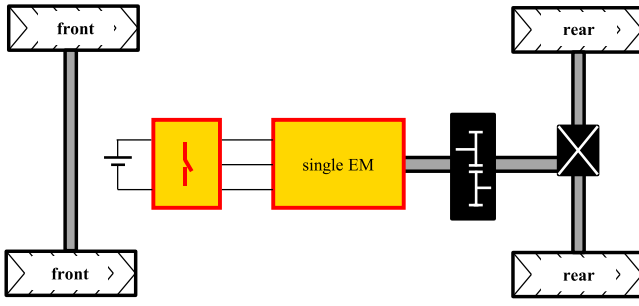


Fig. 1. Configuration of the reference e-bus.

the new topology and its control, rather than from other factors.

- 5) The detailed mechanism behind the energy saving of the proposed topology is analyzed quantitatively and qualitatively.
- 6) The reliability of the proposed four-drive system is demonstrated in comparison to the reference e-bus.
- 7) A sensitivity study shows that the mass increase due to the multidrive system does not affect energy saving.

Section II deals with the modeling and control of a single-drive e-bus as a reference. Section III refers to the modeling and control of the four-drive e-bus. Section IV corresponds to the energy management of the four-drive e-bus. Section V presents the comparison between both e-buses.

II. MODELING AND CONTROL OF THE SINGLE-DRIVE E-BUS

In this article, an ALTAS interurban minibus is used as a case study. It is a battery-powered (400 V) e-bus with a gross mass of 5800 kg. The single electric drive is connected to the rear axle through a final gear (Fig. 1). Due to confidential issues, there is no access to the detailed parameters of the original electric drive.

Thus, a 160-kW three-phase permanent magnet synchronous machine was designed by the finite element method, and its efficiency map is shown in Fig. 2. The EM has a maximal torque of 400 N·m, a maximal speed of 6000 r/min, and a maximal efficiency of 96.5%. Considering that the inverter has a higher average efficiency than the EM, an equivalent efficiency is enough at system-level simulation [15]. In this case, the efficiency of the IGBT-based inverter is considered a constant of 96%. The losses of IGBT, GaN, or SiC-based semiconductors are primarily composed of conduction losses and switching losses. In inverters, the switching frequencies of the semiconductors are constant, with switching losses accounting for most of the total losses. In addition, for the bus cycles, most of the operation points of the EM are along the x -axis (low torque area) due to frequent starts and stops, where efficiencies are typically low. Therefore, assuming a constant inverter efficiency is acceptable for this study. This assumption is validated by our previous experimental tests [15].

A. Modeling and Control

The modeling and control equations of this e-bus are shown in Table I. Equation (1) refers to the battery model with an open-circuit voltage u_{ocv} and internal resistance r_{int} , which are

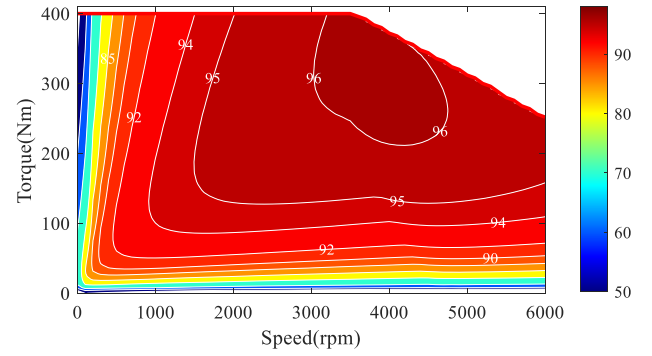


Fig. 2. Efficiency map of the single EM.

TABLE I
MODELING AND CONTROL OF THE SINGLE-DRIVE E-BUS

Modelling	
$u_{bat} = u_{ocv}(SoC) - i_{bat}r_{int}(SoC)$	(1)
$i_{aux} = \frac{P_{aux}}{u_{aux}}$	(2)
$\begin{cases} u_{ed} = u_{aux} = u_{bat} \\ i_{bat} = i_{ed} + i_{aux} \end{cases}$	(3)
$\begin{cases} T_{ed} = T_{ed_ref} \\ i_{ed} = \frac{T_{ed}\Omega_{ed}}{u_{ed}\eta_{ed}} \text{ with } \gamma = \text{sign}(T_{ed}\Omega_{ed}) \end{cases}$	(4)
$\begin{cases} F_{trac} = k_{trans}T_{ed}\eta_{trans}^\lambda \text{ with } \lambda = \text{sign}(F_{wh}v_{ev}) \\ \Omega_{ed} = k_{trans}v_{ev} \end{cases}$	(5)
$\begin{cases} F_{wh} = F_{trac} + F_{bk} \\ v_{ev} = \text{common} \end{cases}$	(6)
$M_{ev} \frac{d}{dt} v_{ev} = F_{wh} - F_{res}$	(7)
$F_{res} = M_{ev}g\sin\alpha + M_{ev}gf\cos\alpha + \sigma v_{ev}^2$	(8)
Control	
$F_{wh_ref} = c_{ev}(v_{ev_ref} - v_{ev_mea}) + F_{res_mea}$	(9)
$F_{trac_ref} = F_{wh_ref} - F_{bk_ref}$	(10)
$T_{ed_ref} = \frac{F_{trac_ref}}{k_{trans}}$	(11)

functions of the SoC, and u_{bat} and i_{bat} are the battery voltage and current, respectively. Equation (2) deduces the current of auxiliary device i_{aux} with a constant power P_{aux} . This current is related to all auxiliaries such as lighting, electronics, and heating ventilation, and air conditioning (HVAC). Equation (3) is the electrical parallel connection, u_{ed} and i_{ed} are the voltage and current of the electric drive, respectively, and u_{aux} is the auxiliary voltage. The single electric drive is described by a static model using (4), where η_{ed} is the efficiency map of the EM and the inverter; γ represents the power flows of the EM; and T_{ed} and Ω_{ed} are the torque and speed, respectively. Equations (5)–(8) correspond to the transmission, chassis, and environment. F_{trac} , F_{wh} , F_{bk} , and F_{res} are forces of the traction system, wheel, braking, and resistance, respectively. λ is a coefficient to distinguish the traction and braking modes. η_{trans} and k_{trans} are, respectively, the efficiency and ratio of the transmission. M_{ev} and v_{ev} are the mass and velocity of the bus, respectively. g is the gravity, f is the rolling coefficient,

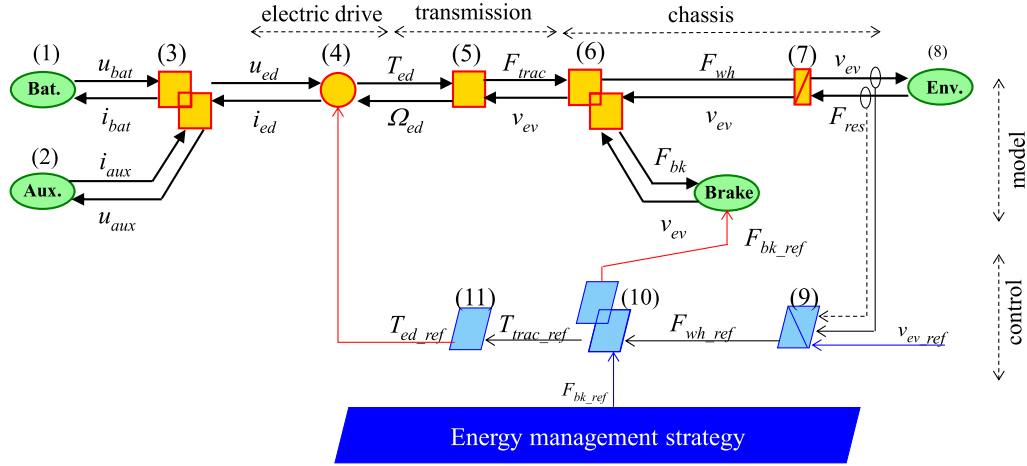


Fig. 3. EMR and control of the single-drive e-bus.

α is the slope angle, and σ is the coefficient for air drag. The model is described using EMR, a graphical description tool for complex models [30]. The EMR of the single-drive e-bus is shown in the upper part of Fig. 3, with corresponding relationships in brackets. The control of the bus is deduced according to inversion rules [31], as indicated in (9)–(11), where c_{ev} is the velocity controller and subscripts “_ref” and “_mea” refer to the reference and measurement, respectively. A control variable F_{bk_ref} for braking strategy is imposed from the energy management strategy, which represents the reference of mechanical braking force. The control of the bus is shown in the lower part of Fig. 3. More details about the modeling and control of a vehicle can be found in [14].

B. Model Validation

As indicated in Fig. 3, the braking of the bus is realized by a control variable F_{bk_ref} . It is shown in [4] that most of the braking force should be provided by the driven axle to keep the stability of the vehicle. Hence, the braking strategy is defined in the following way:

$$F_{bk_ref} = \begin{cases} 0, & (F_{wh} v_{ev} \geq 0) \\ F_{wh_ref}, & (F_{wh} v_{ev} < 0 \text{ \& } v_{ev} \leq 5 \text{ km/h}) \\ 0.4 F_{wh_ref}, & (F_{wh} v_{ev} < 0 \text{ \& } v_{ev} > 5 \text{ km/h}) \end{cases} \quad (12)$$

where F_{bk_ref} is set to zero in the traction mode. In the braking mode, only mechanical force will be used to ensure a reliable deceleration when the velocity is lower than 5 km/h. Otherwise, 40% of the braking force will be provided by mechanical brake disks, and the remaining 60% will be realized by the electric drive from the driven axle in the rear. The work in this article is based on standard driving cycles, and urgent braking is not considered.

The bus is simulated under SORT cycle 2 with an EMR organization using MATLAB/Simulink, resulting in a driving range of 275 km with a 115-kWh battery. The official data for the ALTAS bus under SORT 2 are 270 km, showing a difference of 1.8%. For a system-level simulation, this

TABLE II
MODEL ACCURACY COMPARED TO THE HIGH-FIDELITY
MODEL BY GT SUITE

kWh/100km	New York Bus	London Bus	Denver Bus
GT suite model	63.0	42.4	57.4
EMR-based model	63.9	41.6	57.5
Difference (%)	1.4	1.8	0.2

accuracy is acceptable, proving the reliability of the bus model. We chose MATLAB/Simulink for this study due to its flexibility in implementing and testing control and the energy management strategy (using an EMR library). This allows us to develop a flexible simulation of the four-drive topology and its control in Sections III–V. To ensure the accuracy of our MATLAB/Simulink model, we used the high-fidelity GT-Suite software to model the single-drive e-bus. This GT-Suite model and its validated vehicle library serve as a benchmark to validate the accuracy of the EMR-based model under three real urban bus driving cycles: New York Bus, London Bus, and Denver Bus (Fig. 4). The accuracy in terms of energy consumption is shown in Table II. In the automotive industry, a margin of difference of less than 5% is generally considered acceptable for applications such as fuel consumption predictions, emissions testing, and performance assessments. Our model difference is less than 2%, demonstrating good accuracy [32]. The battery power under New York Bus cycle is taken as an example to illustrate the differences between the two models (Fig. 4), showing that the EMR-based model produces almost the same behavior as the high-fidelity GT model.

III. MODELING AND CONTROL OF THE FOUR-DRIVE E-BUS

The single drive of the original e-bus is replaced by four CE2I drives, where two CE2I drives are located on the front and rear axles (Fig. 5). On each axle, both EMs are connected by sharing the same shafts. The integrated and modular design of the CE2I drive leads to an easy coupling of several drives [29], [33], [34]. Indeed, as the power converter

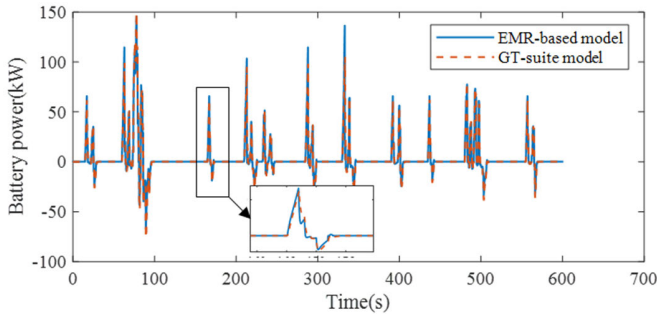


Fig. 4. Comparisons of battery power under New York Bus with two models.

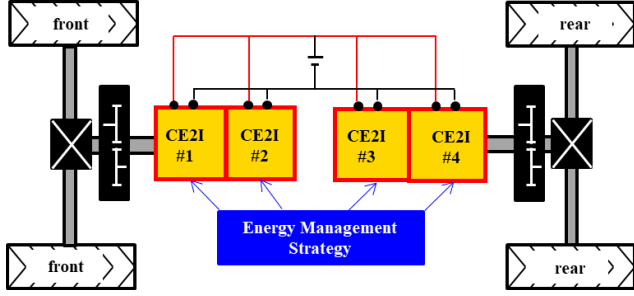


Fig. 5. Configuration of the four-drive e-bus with two driven axles.

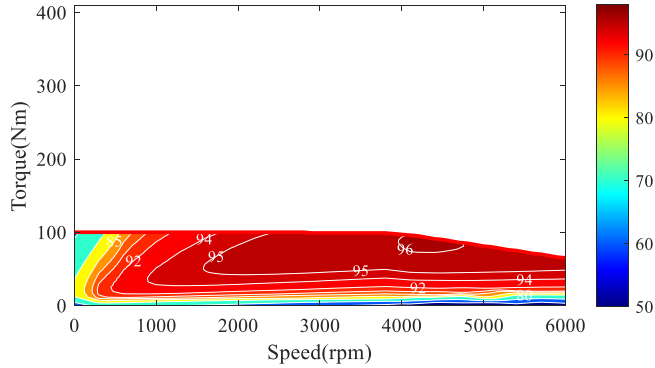


Fig. 6. Efficiency map of the CE2I machine.

is integrated, there are only two dc terminals (electrical ports) and one terminal shaft (mechanical port). The available electric port makes connecting to a unique dc bus more convenient. Compared to the original single-drive e-bus, this four-drive e-bus has an additional mechanical transmission and differential on the front axle. An energy management strategy is required to optimize the power split of each CE2I drive. For a system-level simulation, the efficiency map of the EM is obtained from a finite element method (Fig. 6). The CE2I drive has the same speed range as the single drive (see Fig. 5). Its maximal torque is 1/4 of the single drive, so the 4-CE2I drives have the same maximal torque-speed plane as the single drive. This can guarantee a fair comparison. Moreover, the maximal efficiency of this EM is 96.1% and the equivalent efficiency of the GaN-based inverter is considered as 98%.

A. Modeling of the e-Bus

The modeling and control equations of the four-drive e-bus are shown in Table III. Equation (13) refers to the electrical

TABLE III
MODELING AND CONTROL OF THE FOUR-DRIVE E-BUS

Modelling	
$\begin{cases} u_{ed,r} = u_{ed,f} = u_{ed} \\ i_{ed} = i_{ed,f} + i_{ed,r} \end{cases}$	(13)
$\begin{cases} u_{ed,i} = u_{ed,i+1} = u_{ed,k} \\ i_{ed,k} = i_{ed,i} + i_{ed,i+1} \end{cases}$	(14)
$\begin{cases} T_{ed,k} = T_{ed,i} + T_{ed,i+1} \\ \Omega_{ed,i} = \Omega_{ed,i+1} = \Omega_{ed,k} \end{cases}$	(15)
$\begin{cases} F_{wh} = F_{wh,f} + F_{wh,r} \\ v_{ev,f} = v_{ev,r} = v_{ev} \end{cases}$	(16)
Control	
$F_{wh,f,ref} = F_{wh,ref} - F_{wh,r,ref}$	(17)
$F_{trac,k,ref} = F_{wh,k,ref} - F_{bk,k,ref}$	(18)
$T_{ed,i+1,ref} = T_{ed,k,ref} - T_{ed,i,ref}$	(19)

parallel connection of the front and rear drives, and they share the same dc bus, while the currents are added, where “ $_f$ ” and “ $_r$ ” represent the dual-CE2I drive on the front and the rear, respectively. Similarly, this corresponds to the connection of the two CE2I drives, where subscript $i = [1 \text{ or } 3]$ represents the ID of the CE2I drive and $k = [f \text{ or } r]$. Equation (15) is the mechanical connection of two CE2I drives on each axle. Equation (16) refers to the force coupling of the chassis through both axles. Other relationships are the same as the single-drive e-bus (see Table I). The model of the four-drive e-bus is shown in the upper part of Fig. 7. The control of the e-bus is deduced based on inversion rules. Equation (17) is the control of force coupling of the wheels (16) by introducing a control criteria $F_{wh,f,ref}$. Equation (18) is the inversion of force couplings of each driven axle (6). Equation (19) is the inversion of torque coupling (15). The control of other parts is the same as those in Table I. The control of the four-drive e-bus is shown in the lower part of Fig. 7. There are five control variables: $F_{wh,f,ref}$, $F_{bk,f,ref}$, $F_{bk,r,ref}$, $T_{ed,1,ref}$, and $T_{ed,3,ref}$. Four degrees of freedom are added compared to the original bus.

B. Simulation of the e-Bus With a Simple Strategy

To initially verify the benefits of the power split of the 4-CE2I drives, the e-bus is simulated with a simple strategy

$$F_{bk,k,ref} = \begin{cases} 0, & (F_{wh}v_{ev} \geq 0) \\ F_{wh,k,ref}, & (F_{wh}v_{ev} < 0 \ \& \ v_{ev} \leq 5 \text{ km/h}) \\ 0.4F_{wh,k,ref}, & (F_{wh}v_{ev} < 0 \ \& \ v_{ev} > 5 \text{ km/h}) \end{cases} \quad (20)$$

$$F_{wh,f,ref} = 0.5 F_{wh,ref} \quad (21)$$

$$T_{ed,i,ref} = 0.5 T_{ed,k,ref}. \quad (22)$$

The braking strategy is the same as defined in (12), whereas the power is equally distributed into the 4-CE2I drives. The driving range of the four-drive e-bus is 287 km under SORT

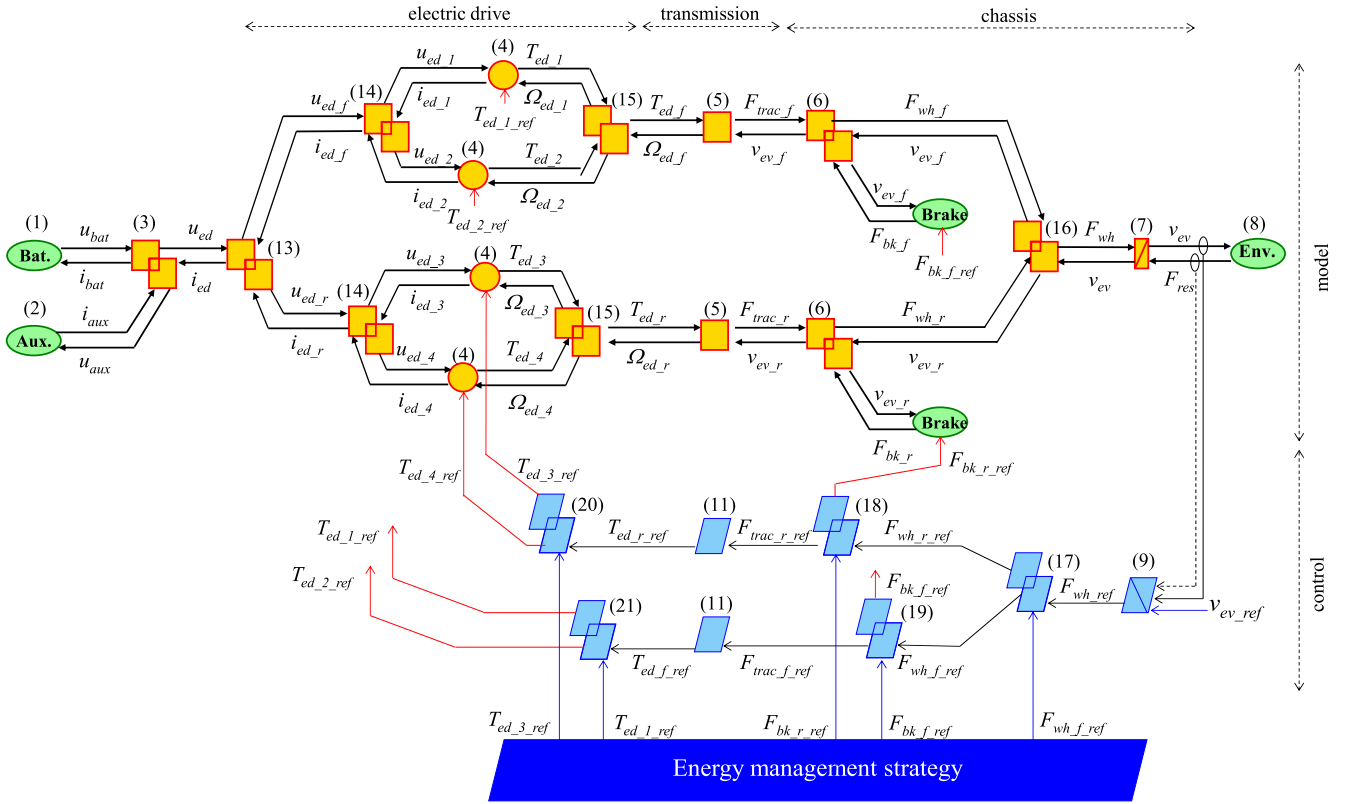


Fig. 7. EMR and control of the four-drive e-bus.

2 with a battery of 115 kWh. Compared to the single-drive e-bus, the driving range is improved by 4.3% (see Section II-B). This simple strategy indicates the efficiency of the new traction topology. Indeed, higher overall system efficiency is expected to be achieved with an optimal energy management strategy.

IV. ENERGY MANAGEMENT OF THE FOUR-DRIVE E-BUSES

In this section, the global efficiency of the entire traction system of the four-drive e-bus is deduced as a function of control variables. Therefore, an optimal energy management strategy is developed to maximize the overall efficiency.

A. Overall Efficiency of the Four-Drive e-Bus

The overall efficiency of the four-drive e-bus should be analyzed to define the optimal energy management strategy. Taking the force reference of the front axle $F_{wh_f_ref}$, mechanical braking force reference of the front axle $F_{bk_f_ref}$, and the torque reference of the 1st CE2I drive $T_{ed_1_ref}$ as control variables, the efficiency of the front axle η_f is given by the following relationship (23), as shown at the bottom of the next page, where $P_{wh_f_ref}$, P_{ed1_ref} , and $P_{bk_f_ref}$ are the powers of the front axle, the 1st CE2I drive, and front mechanical braking, respectively.

In the same way, the efficiency of the rear axle is given by (24), as shown at the bottom of the next page.

The overall efficiency of the four-drive e-bus is described in a function of η_f and η_r

$$\eta_{sys} = \left(\frac{P_{wh}}{P_{ed_f_elec} + P_{ed_r_elec}} \right)^\lambda = \left(\frac{F_{wh_ref}}{F_{wh_f_ref}\eta_f^{-\lambda} + (F_{wh_ref} - F_{wh_f_ref})\eta_r^{-\lambda}} \right)^\lambda. \quad (25)$$

In this way, the overall efficiency η_{sys} is expressed as a function of the control variables: $F_{wh_r_ref}$, $F_{bk_f_ref}$, $F_{bk_r_ref}$, $T_{ed_1_ref}$, and $T_{ed_3_ref}$.

B. Optimal Energy Management Strategy

The main philosophy of the optimal energy management strategy is to define the value of $F_{wh_r_ref}$, $F_{bk_f_ref}$, $F_{bk_r_ref}$, $T_{ed_1_ref}$, and $T_{ed_3_ref}$ for a certain power demand with F_{wh_ref} and v_{ev} , to maximize η_{sys} based on the efficiency maps of the 4-CE2I drives (Fig. 8).

The studied four-drive e-bus has two CE2I drives on both driven axles, so both axles can recover energy during braking. In this way, the braking strategy is defined as follows:

$$F_{bk_k_ref} = \begin{cases} 0, & (F_{wh}v_{ev} \geq 0) \\ F_{wh_k_ref}, & (F_{wh}v_{ev} < 0 \text{ \& } v_{ev} \leq 5 \text{ km/h}) \\ 0, & (F_{wh}v_{ev} < 0 \text{ \& } v_{ev} > 5 \text{ km/h}). \end{cases} \quad (26)$$

It indicates that pure mechanical braking is only used when the velocity is below 5 km/h. Otherwise, pure electric braking

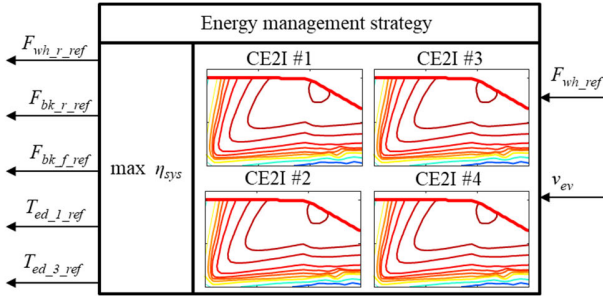


Fig. 8. Philosophy of the energy management strategy.

is activated. This rule ensures optimal braking while also respecting the braking safety requirements [35].

Moreover, the distribution of torques between the four drives should be defined to increase global efficiency. Meanwhile, the 4-CE2I drives should respect the corresponding physical constraints to meet the power demand. Therefore, the optimization problem aims to maximize the global efficiency

$$\max_{\mu} \eta_{sys}(\mu) \quad (27)$$

by acting on the control variable

$$\mu = [F_{wh_f_ref}, F_{bk_f_ref}, F_{bk_r_ref}, T_{ed_1_ref}, T_{ed_3_ref}] \quad (28)$$

respecting the constraints

$$T_{ed_s_min} \leq T_{ed_s} \leq T_{ed_s_max} \quad (29)$$

with

$$T_{ed_s_max} = \begin{cases} T_{ed_max}, & (\Omega_s \leq \Omega_N) \\ \frac{T_{ed_max}\Omega_N}{\Omega_s}, & (\Omega_N < \Omega_s \leq \Omega_{max}) \end{cases} \quad (30)$$

where subscript $s = [1], [2], [3], [4]$ refers to the CE2I ID and “ $_{-max}$ ” and “ $_N$ ” correspond to the maximal and rated items, respectively.

The optimization problem is solved by a brute-force search method (Fig. 9).

Step 1: The total force demand F_{wh_ref} at the wheel side and the actual bus velocity v_{ev} are obtained based on the EMR and control defined in Fig. 7.

Step 2: All the control variables are initialized, the initial overall efficiency η_{sys} is set as 0, and the mechanical braking forces are given by (26).

Step 3: Check all the constraints defined by (29) and (30).

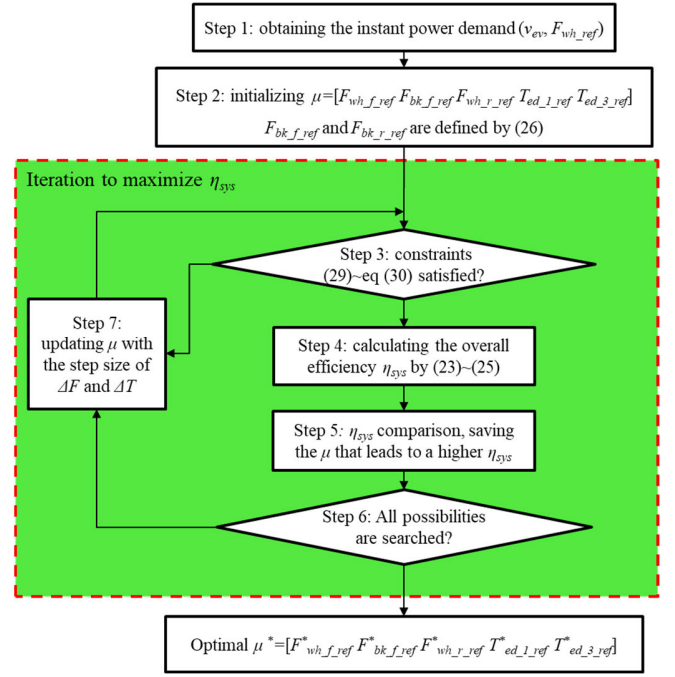


Fig. 9. Flowchart of the optimization problem.

Step 4: Obtain the efficiencies of the 4-CE2I drives by lookup table, and then, the system efficiency η_{sys} is calculated by (23)–(25).

Step 5: Compare the saved efficiency in the buffer with the overall efficiency derived in Step 4, and select the higher efficiency and the respective control variables.

Step 6: Check if all the feasible possibilities are searched with a certain step size.

Step 7: Update the control variables with the step of ΔF and ΔT ($\Delta F = 50$ N and $\Delta T = 1$ N·m). Please note that if the step sizes ΔF and ΔT are small enough, the overall efficiency is optimal. It will search all the possibilities in the feasible region, and thereby, the strategy is globally optimal.

V. COMPARISONS OF BOTH E-BUSES

The energy-saving capability of the four-drive e-bus is emphasized through a comparison with the original bus under real bus driving cycles. The mechanism behind this energy saving is analyzed in detail. The reliability of the four-drive system is simulated in the fault mode when one drive is

$$\begin{aligned} \eta_f &= \left(\frac{P_{wh_f_ref}}{P_{ed1_ref}\eta_{ed1}^{-\gamma} + [(P_{wh_f_ref} - P_{bk_f_ref})\eta_{trans}^{-\lambda} - P_{ed1_ref}]\eta_{ed2}^{-\gamma}} \right)^{\lambda} \\ &= \left(\frac{F_{wh_f_ref}k_{trans}}{T_{ed1_ref}\eta_{ed1}^{-\gamma} + k_{trans}[(F_{wh_f_ref} - F_{bk_f_ref})\eta_{trans}^{-\lambda} - T_{ed1_ref}]\eta_{ed2}^{-\gamma}} \right)^{\lambda} \end{aligned} \quad (23)$$

$$\eta_r = \left(\frac{(F_{wh_ref} - F_{wh_f_ref})k_{trans}}{T_{ed3_ref}\eta_{ed3}^{-\gamma} + k_{trans}[(F_{wh_ref} - F_{wh_f_ref} - F_{bk_r_ref})\eta_{trans}^{-\lambda} - T_{ed3_ref}]\eta_{ed4}^{-\gamma}} \right)^{\lambda} \quad (24)$$

TABLE IV
PARAMETERS OF DIFFERENT BUS CYCLES

Parameters	New York Bus	London Bus	Denver Bus
Average velocity (km/h)	5.93	13.27	7.24
Maximal velocity (km/h)	49.57	41.96	32.69
Maximal acceleration (m/s^2)	2.77	1.31	1.7
Maximal deceleration (m/s^2)	-2.06	-1.66	-2.27
Energy ratio: E_{bk}/E_{trac} (%)	75	60	74

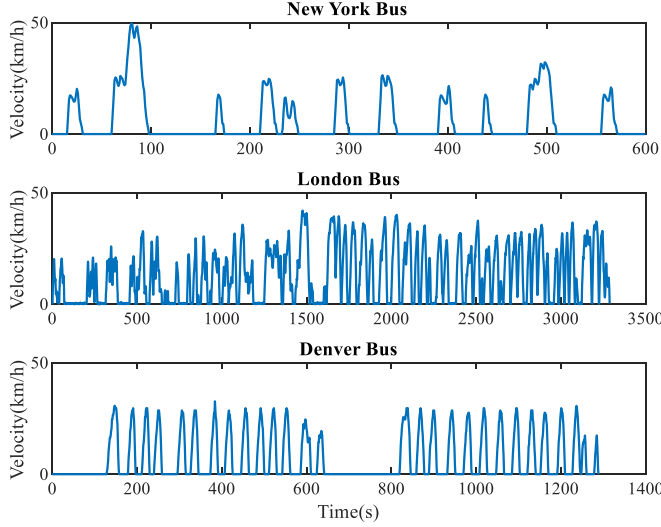


Fig. 10. Studied driving cycles.

compromised. A comprehensive comparison of both e-buses is carried out and a sensitivity analysis is performed to understand the impact of the four-drive mass on energy savings.

Three real urban bus driving cycles are used for a more detailed comparison, i.e., New York Bus, London Bus, and Denver Bus (Fig. 10). There are frequent accelerations and decelerations, while fewer high-speed cruising in these urban bus cycles. Table IV shows some key properties of these cycles. The ratio of braking energy to traction energy is up to 75%, which is much higher than that of car cycles. This indicates that buses have greater potential to recover energy during braking.

A. Comparisons in Normal Operation

Based on the algorithm in Fig. 9, the optimal energy management strategy is calculated off-line. Then, the strategy is implemented using lookup tables. Compared to the single-drive e-bus, the proposed four-drive e-bus saves energy by up to 32% under the studied cycles (Table V). This means that the driving range of the bus can be extended by 32% with the same battery, thanks to the proposed topology with 4-CE2I drives. An evaluated gross mass increment of the four-drive e-bus is considered in the simulation. This includes the mass of an additional final gearbox, as well as the mass increment of 4-CE2I drives in comparison to the single drive.

To better explain the significant energy saving, Fig. 11 shows the energy flow of the studied e-bus. In the traction mode (the yellow paths), E_{aux} and E_{pt_trac} are the energy consumed by the auxiliary and powertrain (electric drive + transmission), respectively. The rest of the energy balances

TABLE V
COMPARISON OF ENERGY CONSUMPTION OF BOTH E-BUSES

kWh/100km	SORT2	New York Bus	London Bus	Denver Bus
Single-drive e-bus	41.8	63.9	41.6	57.5
Four-drive e-bus	32.3	44.0	30.2	39.2
Energy saving (%)	22.7	31.3	27.4	31.8

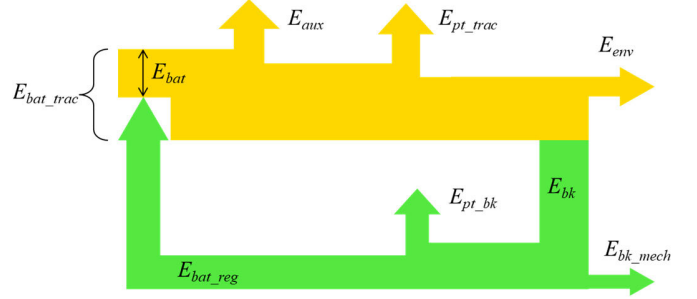


Fig. 11. Energy flow of the studied e-bus.

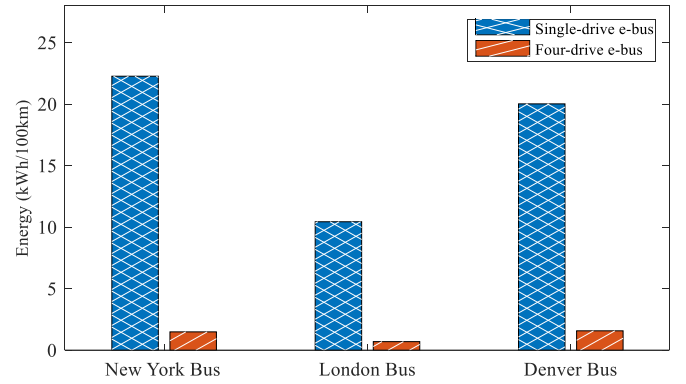
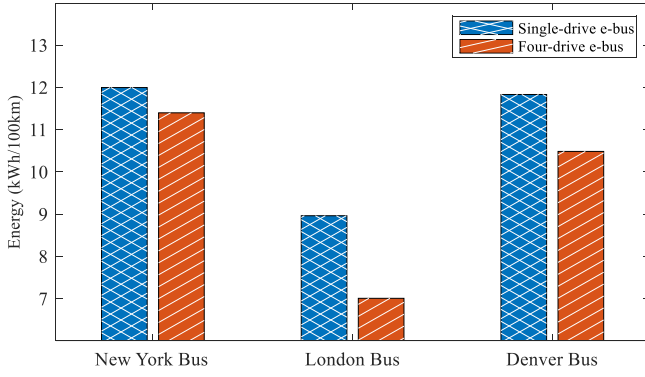


Fig. 12. Mechanical braking energy of the two e-buses (E_{bk_mech}).

the energy loss E_{env} from the environment and braking energy E_{bk} . In the braking mode (green paths), E_{bk_mech} is the lost energy due to mechanical brake disks and E_{pt_bk} is the energy loss of the powertrain. The rest of the energy E_{bat_reg} can be recovered by the battery. The final consumed energy is E_{bat} .

For a given cycle, E_{aux} and E_{env} are constant when the bus tracks the reference. For the single-drive e-bus, an average braking distribution coefficient is assumed as 40% when the velocity is higher than 5 km/h, and this value has been verified in the simulation in Section II. Nevertheless, for the four-drive e-bus, both driven axles can recover energy during braking so that the braking coefficient is zero. This means that 40% more of the braking energy is recovered. Fig. 12 shows the amount of wasted energy due to mechanical braking under the studied cycles. The wasted energy in the single-drive e-bus is much higher. The four-drive e-bus also has some braking energy loss since pure mechanical braking is enabled when the velocity is lower than 5 km/h to ensure reliable deceleration. Moreover, in urban bus driving cycles, there are frequent acceleration and deceleration, and the average velocity is relatively low; thus, E_{bk} is much higher than E_{env} . Therefore, the impact of energy recovery is more important in comparison to other driving cycles.

In addition to the braking coefficient, the powertrain losses E_{pt_trac} and E_{pt_bk} also influence energy consumption. Considering that both e-buses have different regenerative power

Fig. 13. Energy loss of the powertrains in traction mode (E_{pt_trac}).

during braking, Fig. 13 shows the comparisons of powertrain losses in traction. It indicates that the four-drive e-bus wastes less energy in the powertrain in the traction mode. This improvement benefits from the optimal power split between each CE2I drive, even though the maximal efficiency of the CE2I machine is a little bit lower than that of the single-drive EM.

Regarding the entire traction system, an average overall efficiency $\bar{\eta}_{sys}$ is calculated for comparison

$$\bar{\eta}_{sys} = \frac{\sum_{\kappa=0}^L F_{wh}(\kappa) v_{ev}(\kappa) \Delta t}{\sum_{\kappa=0}^L u_{ed}(\kappa) i_{ed}(\kappa) \Delta t} \quad (31)$$

where L is the total length of the cycle and Δt is the step size.

Fig. 14 shows the average efficiencies of the powertrain of both e-buses in traction mode. The average efficiency of the CE2I-based drive is around 88%, which is higher than that of the single drive. The maximal efficiency improvement is observed in the cycle of the London Bus, which is 6.5%. Compared to the single-drive e-bus, the average efficiency of the CE2I-based drive changes less under different cycles. This is due to the better optimization allowed by the additional degrees of freedom. However, it is impossible to optimize the operating points of the single drive, and hence, its efficiency varies with different driving conditions. For example, for the power demand at (3000 r/min, 140 N·m), the efficiency of the single drive is 85% (including the efficiency of the inverter). The four-drive system has an efficiency of 93.6% (including the efficiency of the inverter) with the optimal torque distribution of 70, 70, 0, and 0 N·m to the four drives. At this power demand, the efficiency of the traction system is improved by 8.6%. The results obtained in this article are based on one case where the 4-CE2I drives are identical. Indeed, better energy efficiency could be realized by optimally sizing the powers of the four electric drives.

The overall battery-to-wheel efficiency of the four-drive e-buses is shown in Fig. 15. This figure is derived using the efficiency maps of the 4-CE2I drives, considering the mechanical transmission system parameters and the energy management strategy defined in Fig. 9. In traction (positive force), the maximal overall efficiency of the four-drive e-buses reaches up to 92%. For velocities higher than 30 km/h, there are four dark-colored areas corresponding to

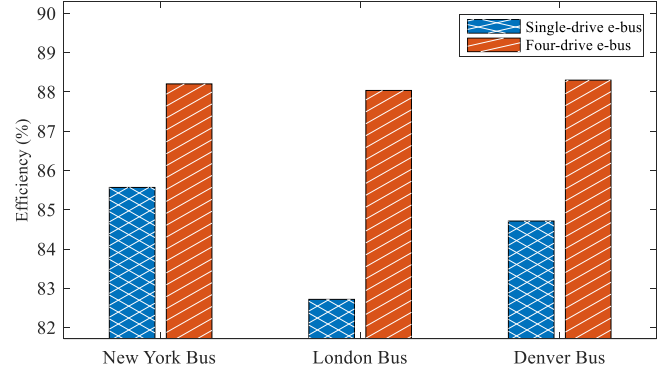


Fig. 14. Average efficiencies of the powertrains in traction mode.

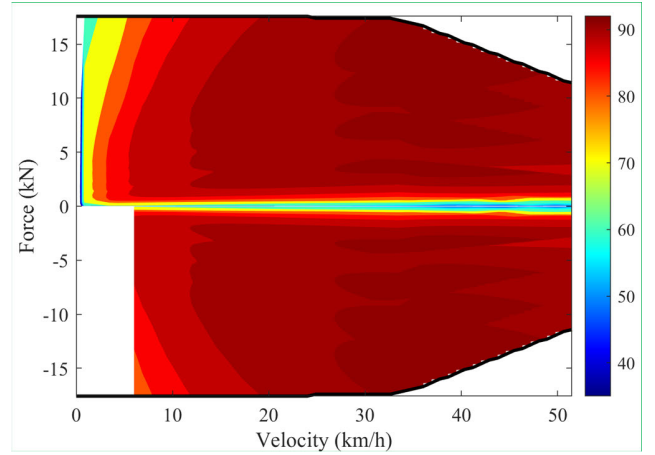


Fig. 15. Battery-to-wheel efficiency map of the four-drive e-bus.

the high-efficiency zones of the 4-CE2I drives. In braking (negative force), no energy is recovered when the velocity is below 5 km/h. However, when the velocity exceeds 5 km/h, the 4-CE2I drives recover energy with the best efficiencies. For comparison, the overall battery-to-wheel efficiency of the single-drive e-bus is also shown in Fig. 16. In traction, its maximal efficiency is about 89%, which is lower than that of the four-drive e-bus due to the optimized strategy of the latter. In braking, like the five-drive e-bus, only mechanical force is used to brake the vehicle when the velocity is below 5 km/h. However, when the velocity is higher than 5 km/h, 40% of the braking energy is consumed by mechanical force [see (12)], resulting in a maximal energy recovery efficiency of about 54%, significantly lower than the 92% of the four-drive e-bus. These two figures clearly illustrate the energy-saving capabilities of both e-buses.

B. Comparisons in Fault Operation

In addition to normal operation, the proposed four-drive traction system is expected to be more reliable in the event of a fault. As the 4-CE2I drives are independent of each other, when one breaks down, the others can still operate. Fig. 17 illustrates an example of the e-bus velocity profile under the cycle of the New York Bus. During the normal operation, the real velocity (red dotted line) closely tracks the reference (blue solid line). When one CE2I drive is deactivated, the faulty e-bus (orange dashed line) experiences a slight loss

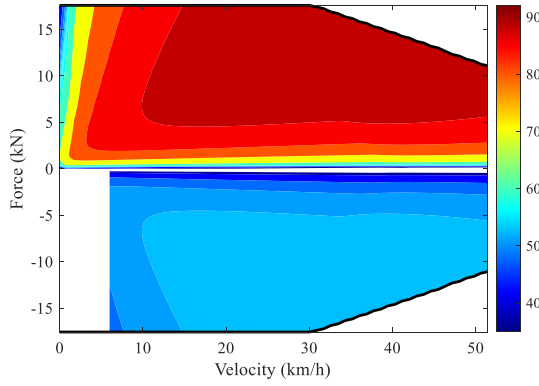


Fig. 16. Battery-to-wheel efficiency map of the single-drive e-bus.

TABLE VI

GAIN OF ENERGY CONSUMPTION OF THE FOUR-DRIVE E-BUS IN FAULT MODE IN COMPARISON TO THE NORMAL MODE

Energy difference	New York Bus	London Bus	Denver Bus
Normal mode	-	-	-
One drive at fault	+1.00%	+0.04%	+0.34%
Two drives at fault	+4.56%	+0.23%	+4.75%

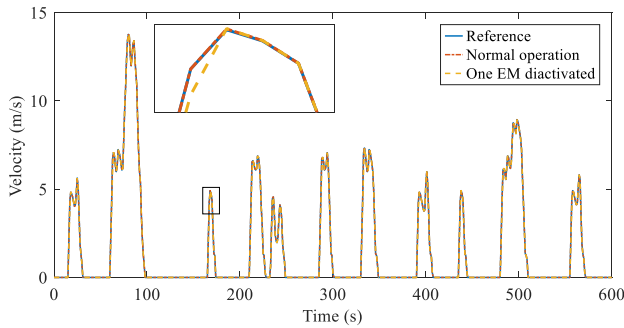


Fig. 17. Velocity profiles under normal operation and fault mode.

of dynamics, but overall, it can still follow the reference. This ensures that the e-bus can proceed to the garage for repair. Table VI presents the difference in terms of energy consumption in the fault mode compared to that of the normal mode. It can be observed that when only one CE2I drive is at fault, the energy consumption is very close to the normal mode, which takes benefit of CE2I efficiency optimization with the optimal strategy. However, when two CE2I drives are at fault, the energy consumption can increase by up to 4.75%. In practice, the likelihood of two CE2I drives breaking down simultaneously is lower than that of only one breaking down. Therefore, when one CE2I drive is at fault, the e-bus should be taken to the garage for repair instead of continuing to operate with a fault.

C. Comparisons With Two Emulated Multimachine Topologies

The proposed four-drive system combines features of a distributed and a centralized multimachine (CM) system. To highlight the benefits of the combination of the two, these two topologies are compared. Thanks to a flexible control organization using EMR, a distributed multimachine (DM)

TABLE VII

COMPARISONS OF THE THREE TOPOLOGIES

(%)	New York Bus			London Bus			Denver Bus		
$\bar{\eta}_{sys_trac}$	R	DM	CM	R	DM	CM	R	DM	CM
$\bar{\eta}_{sys_bk}$	88.2	87.8	88.2	88.1	87.1	88.0	88.3	87.5	88.3
	83.8	82.7	50.0	85.7	84.1	76.0	85.5	84.5	66.2

system or a CM system can be emulated by managing the five control signals imposed from the energy management strategy (Fig. 7) without system reconfiguration.

- 1) *Emulation of a DM System:* The forces on the front and rear axles are equally distributed during traction and braking phases ($F_{wh,r_ref} = 0.5F_{wh_ref}$). The other four control signals are allocated using an optimal power split strategy defined in Fig. 9. In this mode, the system is equivalent to a two-drive-axle system with one equivalent drive on the front and one on the rear. This configuration allows for greater energy recovery during braking compared to a single drive. However, the force distribution between the axles is not optimal.
- 2) *Emulation of a CM System:* The main idea is to reproduce a multimachine on a single axle. To keep the same power, the 4-CE2I drives are considered in the front axle and no drives are in the rear. In the traction mode, the power is optimally distributed between all the drives. In the braking mode, 40% is insured by the electric drives and 60% is insured by mechanical braking.

For the proposed topology, a combination of a distributed and a CM system is taken as a reference (R). The five control signals from the energy management strategy (forces on the front and rear axles, as well as the torques among the 4-CE2I drives) are optimally distributed according to the strategy outlined in Fig. 9. This control offers the greatest potential for energy savings.

Using (31), the overall efficiencies of the four-drive e-bus in traction ($\bar{\eta}_{sys_trac}$) and braking ($\bar{\eta}_{sys_bk}$) with different topologies and driving cycles are shown in Table VII.

The reference topology (R) exhibits the highest average efficiencies in both traction and braking phases. The high braking efficiency benefits from energy recovery of both driven axles (distributed system). The average traction efficiencies are higher than those of a single drive (see plots in Fig. 14). This is due to efficiency improvement of the two cascaded CE2I drives (centralized system) on both axles managed by an optimal torque distribution strategy (see Fig. 9). In the emulated DM system, the efficiencies are slightly lower than in the reference (R), as the forces are equally distributed between the two axles. The comparison between DM and R shows that the gain comes from the power split optimization between two drive axles. The emulated CM system, equivalent to having multiple drives on the front axle, achieves high average traction efficiency similar to the reference (R). However, since only one axle recovers braking energy, a significant amount of energy is wasted. The comparison between CM and R demonstrates that the improvement is ensured by recovering braking energy on both axles. The EMR-based control organization allows for a comparison of different multimachine topologies.

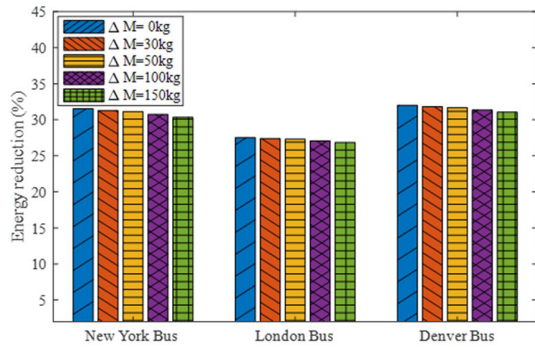


Fig. 18. Sensitivity analysis on the impact of system mass increases on energy reductions under different driving conditions.

D. Sensitivity Analysis on Global Mass Impacts

Switching from a single-drive system to the proposed four-drive system increases the traction total mass because the system power density decreases with the addition of more auxiliary materials [16]. Auxiliary materials refer to the nonactive components that are necessary for the operation and structural integrity of the system but do not directly contribute to its power output. At the initial feasibility study phase, it is difficult to have a specific value for the mass increase, and hence, this section performs a sensitivity analysis. Assuming mass increases of 0, 30, 50, 100, and 150 kg, simulations comparing energy reductions to a single-drive e-bus are conducted (as shown in Fig. 18). The results indicate that under favorable conditions where the system mass remains unchanged ($\Delta m = 0$ kg, represented by the blue bar), the four-drive e-bus achieves energy saving rates of 31.5%, 27.5%, and 32.0% compared to the single-drive e-bus across the three cycles. In an extreme scenario where the system mass increases by 150 kg (the green bar), the energy-saving rates are 30.3%, 26.8%, and 31.0%. These findings demonstrate that the benefits gained from the proposed four-drive system, coupled with an optimal energy management strategy, far exceed the energy loss resulting from the increase in system mass.

The results in Fig. 18 can be attributed to two main factors. First, considering the substantial mass of the e-bus (5800 kg), an increment of 150 kg represents only a marginal increase of up to 2.5% in the overall mass, which is relatively negligible to some extent. Second, the optimal energy management strategy defined in Fig. 9 ensures that the system efficiency remains at its optimum even under disturbances, such as changes in mass due to variations in passenger numbers.

As for the capital cost, it is quite difficult to compare quantitatively, as the price of the materials would vary with time and different suppliers. Readers who are interested in this could refer to our previous work in [16], where there is a capital cost analysis for multidrive systems. Even though the capital cost will increase, the energy savings will pay back the capital cost after several years. If the yearly driving range of an e-bus is D_y , the price of electricity is ρ_{ele} , the energy saving rate and energy consumption under a certain cycle is r and E_{bat} , and the driving range of a cycle is D_c , then the yearly saved energy cost is

$$c_e = \frac{D_y}{D_c} r E_{bat} \rho_{ele}. \quad (32)$$

TABLE VIII
YEARLY SAVED ENERGY COST OF THE FOUR-DRIVE E-BUS
FOR 300 000 KM

	New York Bus	London Bus	Denver Bus
Saved cost (€)	922	522	837

For the three cycles selected in this article, suppose that $D_y = 30\,000$ km and $\rho_{ele} = 0.15$ e/kWh so that the yearly saved energy costs of the three cycles are shown in Table VIII.

Overall, the proposed four-drive system features a configuration with two cascaded drives on each of the front and rear axles. This topology offers several advantages.

- 1) Enhanced efficiency through optimized power distribution among the four drives.
- 2) Increased energy recovery during braking across both driven axles.

However, this configuration also introduces certain challenges.

- 1) The necessity to manage four drives elevates system complexity.
- 2) A reduction in power density and an increase in costs compared to the single-drive setup.

In addition, an EMR-based control framework offers a flexible approach to managing the operation modes of the four-drive system. This method allows for the emulation of different multimachine configurations by adjusting the control signals derived from the energy management strategy, without the need to reconfigure the physical system.

VI. CONCLUSION

A four-drive traction topology with two cascaded smart integrated drives on both axles is proposed for an urban e-bus. The model and control of the reference bus are verified. An optimal energy management strategy is developed to minimize the energy consumption of the four-drive e-bus.

The traction systems of both e-buses have the same total powers and torque-speed planes. Through the simulation under different urban bus driving cycles, an energy-saving rate of up to 32% can be obtained by using the proposed two-drive-axle system with four smart integrated drives. This energy reduction benefits from the following aspects.

- 1) The overall efficiency of the traction system is improved as more degrees of freedom are introduced. A maximal efficiency improvement of 6.5% is observed under the London Bus in the traction mode.
- 2) Both smart integrated drives are located on the front and rear axle, which enables better energy recovery ability.
- 3) An appropriated control based on EMR to take benefit of this topology.
- 4) An optimal energy management strategy maximizes the global efficiency.

The study highlights the interest in applying this new multidrive system to an urban electric bus. However, this topology requires a reconfiguration of both axles and will have higher mass (or volume) and cost. Except for normal operation, the four-drive system demonstrates robust fault tolerance capabilities, maintaining operational integrity even in scenarios where one or two drives encounter faults. Sensitivity analysis

reveals that the increase in gross mass due to the four-drive system has a negligible effect on the energy reduction benefits. This underscores that the gains from the new topology and optimal energy management far outweigh the losses due to mass increase. Furthermore, comparisons with an equivalent DM system and an equivalent CM system quantify the benefits of combining the two topologies.

REFERENCES

- [1] F. Rodríguez. (2018). *Fuel Consumption Simulation of HDVs in the EU: Comparisons and Limitations*. Int. Council Clean Transp. [Online]. Available: https://theicct.org/wp-content/uploads/2021/06/VECTO_Simulation_Comparison_final_20180308.pdf
- [2] C. Buysse, B. Sharpe, and O. Delgado. (2021). *Efficiency Technology Potential for Heavy-Duty Diesel Vehicles in the United States Through 2035*. Int. Council Clean Transp. [Online]. Available: <https://theicct.org/wp-content/uploads/2021/12/efficiency-tech-potential-hdvs-us-2035-nov21.pdf>
- [3] A. Sen and J. Miller. (2023). *Vision 2050: Updata on the Global Zero-Emission Vehicle Transition in 2023*. Int. Council Clean Transp. [Online]. Available: https://theicct.org/wp-content/uploads/2023/09/Global-ZEV-update_final.pdf
- [4] M. Ehsani, Y. Gao, S. Longo, and K. M. Ebrahimi, *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*. Boca Raton, FL, USA: CRC Press, 2018.
- [5] U. R. Muduli, A. R. Beig, K. A. Jaafari, J. Y. Alsawalhi, and R. K. Behera, "Interrupt-free operation of dual-motor four-wheel drive electric vehicle under inverter failure," *IEEE Trans. Transport. Electrification*, vol. 7, no. 1, pp. 329–338, Mar. 2021.
- [6] Z. Wang, J. Zhou, and G. Rizzoni, "A review of architectures and control strategies of dual-motor coupling powertrain systems for battery electric vehicles," *Renew. Sustain. Energy Rev.*, vol. 162, Jul. 2022, Art. no. 112455.
- [7] J. Hong, J. Wu, N. Zhang, B. Wang, and Y. Tian, "The dynamic and economic performance study of a new simpson planetary gearset based dual motor powertrain for electric vehicles," *Mechanism Mach. Theory*, vol. 167, Jan. 2022, Art. no. 104579.
- [8] J. Gong, H. Zahr, E. Semail, M. Trabelsi, B. Aslan, and F. Sculler, "Design considerations of five-phase machine with double p/3p polarity," *IEEE Trans. Energy Convers.*, vol. 34, no. 1, pp. 12–24, Mar. 2019.
- [9] S. Sharma, M. V. Aware, and A. Bhowate, "Symmetrical six-phase induction motor-based integrated driveline of electric vehicle with predictive control," *IEEE Trans. Transport. Electrification*, vol. 6, no. 2, pp. 635–646, Jun. 2020.
- [10] M. A. Frikha, J. Croonen, K. Deepak, Y. Benômar, M. El Baghdadi, and O. Hegazy, "Multiphase motors and drive systems for electric vehicle powertrains: State of the art analysis and future trends," *Energies*, vol. 16, no. 2, p. 768, 2023.
- [11] Z. Yu, C. Gan, K. Ni, Y. Chen, and R. Qu, "Dual-electric-port bidirectional flux-modulated switched reluctance machine drive with multiple charging functions for electric vehicle applications," *IEEE Trans. Power Electron.*, vol. 36, no. 5, pp. 5818–5831, May 2021.
- [12] W. Yan, H. Chen, S. Liao, Y. Liu, and H. Cheng, "Design of a low-ripple double-modular-stator switched reluctance machine for electric vehicle applications," *IEEE Trans. Transport. Electrification*, vol. 7, no. 3, pp. 1349–1358, Sep. 2021.
- [13] H. Chen, A. M. El-Refai, Y. Zuo, S. Cai, L. Cao, and C. H. T. Lee, "Comparative study and design optimization of a dual-mechanical-port electric machine for hybrid electric vehicle applications," *IEEE Trans. Veh. Technol.*, vol. 71, no. 8, pp. 8341–8353, Aug. 2022.
- [14] K. Li, A. Bouscayrol, S. Han, and S. Cui, "Comparisons of electric vehicles using modular cascade machines system and classical single drive electric machine," *IEEE Trans. Veh. Technol.*, vol. 67, no. 1, pp. 354–361, Jan. 2018.
- [15] K. Li, S. Han, S. Cui, and A. Bouscayrol, "Sizing of modular cascade machines system for electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1278–1287, Feb. 2019.
- [16] K. Li, A. Bouscayrol, S. Cui, and Y. Cheng, "A hybrid modular cascade machines system for electric vehicles using induction machine and permanent magnet synchronous machine," *IEEE Trans. Veh. Technol.*, vol. 70, no. 1, pp. 273–281, Jan. 2021.
- [17] K. Kwon, M. Seo, and S. Min, "Efficient multi-objective optimization of gear ratios and motor torque distribution for electric vehicles with two-motor and two-speed powertrain system," *Appl. Energy*, vol. 259, Feb. 2020, Art. no. 114190.
- [18] C. T. Nguyen, P. D. Walker, S. Zhou, and N. Zhang, "Optimal sizing and energy management of an electric vehicle powertrain equipped with two motors and multi-gear ratios," *Mechanism Mach. Theory*, vol. 167, Jan. 2022, Art. no. 104513.
- [19] X. Lin, Y. Li, and G. Zhang, "Bi-objective optimization strategy of energy consumption and shift shock based driving cycle-aware bias coefficients for a novel dual-motor electric vehicle," *Energy*, vol. 249, Jun. 2022, Art. no. 123596.
- [20] C. Zhang, S. Zhang, G. Han, and H. Liu, "Power management comparison for a dual-motor-propulsion system used in a battery electric bus," *IEEE Trans. Ind. Electron.*, vol. 64, no. 5, pp. 3873–3882, May 2017.
- [21] J. Wu and N. Zhang, "Driving mode shift control for planetary gear based dual motor powertrain in electric vehicles," *Mechanism Mach. Theory*, vol. 158, Apr. 2021, Art. no. 104217.
- [22] M. Hu, S. Chen, and J. Zeng, "Control strategy for the mode switch of a novel dual-motor coupling powertrain," *IEEE Trans. Veh. Technol.*, vol. 67, no. 3, pp. 2001–2013, Mar. 2018.
- [23] R. K. Subroto, C. Z. Wang, and K. L. Lian, "Four-wheel independent drive electric vehicle stability control using novel adaptive sliding mode control," *IEEE Trans. Ind. Appl.*, vol. 56, no. 5, pp. 5995–6006, Sep. 2020.
- [24] M. Oh and I. Husain, "Optimal torque distribution of dual-motor all-wheel drive electric vehicles for maximizing motor energy efficiency," in *Proc. IEEE Transp. Electrification Conf. Expo (ITEC)*, Jun. 2021, pp. 188–193.
- [25] D.-M. Kim, P. Benoliel, D.-K. Kim, T. H. Lee, J. W. Park, and J.-P. Hong, "Framework development of series hybrid powertrain design for heavy-duty vehicle considering driving conditions," *IEEE Trans. Veh. Technol.*, vol. 68, no. 7, pp. 6468–6480, Jul. 2019.
- [26] X. Sun, Z. Shi, Y. Cai, G. Lei, Y. Guo, and J. Zhu, "Driving-cycle-oriented design optimization of a permanent magnet hub motor drive system for a four-wheel-drive electric vehicle," *IEEE Trans. Transport. Electrification*, vol. 6, no. 3, pp. 1115–1125, Sep. 2020.
- [27] Z. Zhao, H. Taghavifar, H. Du, Y. Qin, M. Dong, and L. Gu, "In-wheel motor vibration control for distributed-driven electric vehicles: A review," *IEEE Trans. Transport. Electrification*, vol. 7, no. 4, pp. 2864–2880, Dec. 2021.
- [28] J. Hu, J. Li, Z. Hu, B. Zhang, L. Xu, and M. Ouyang, "Energy-efficient torque-allocation strategy for a 6×6 vehicle using electric wheels," *eTransportation*, vol. 10, Nov. 2021, Art. no. 100136.
- [29] B. Lemaire-Semail, N. Idir, E. Semail, and S. Harmand, "Integrated motor drive: A multidisciplinary approach," in *Proc. 24th Eur. Conf. Power Electron. Appl. (EPE ECCE Europe)*, Sep. 2022, pp. P.1–P.9.
- [30] A. Bouscayrol, J. P. Hautier, and B. Lemaire-Semail, "Graphic formalisms for the control of multi-physical energetic systems: COG and EMR," in *Systemic Design Methodologies for Electrical Energy Systems: Analysis, Synthesis and Management*. Hoboken, NJ, USA: Wiley, 2012, pp. 89–124, doi: [10.1002/9781118569863.ch3](https://doi.org/10.1002/9781118569863.ch3).
- [31] A. Bouscayrol and B. Lemaire-Semail, "Energetic macroscopic representation and inversion-based control," in *Encyclopedia of Electrical and Electronic Power Engineering*, J. García, Ed., Oxford, U.K.: Elsevier, Jan. 2023, pp. 365–375, doi: [10.1016/B978-0-12-821204-2.000117-3](https://doi.org/10.1016/B978-0-12-821204-2.000117-3).
- [32] C. Mayet et al., "Comparison of different models and simulation approaches for the energetic study of a subway," *IEEE Trans. Veh. Technol.*, vol. 63, no. 2, pp. 556–565, Feb. 2014.
- [33] S. Eric, I. Nadir, H. Souad, and L.-S. Betty, "Integrated drive and reliabilities: Fault tolerant architectures and supply," presented at the Towards Sustain. Aviation Summit Symp., Toulouse, France, 2022.
- [34] E. Semail, S. Harmand, N. Idir, and B. Lemaire-Semail, "Integrated polyphase electric machine," Eur. Patent WO 202 213 680, Jun. 30, 2022.
- [35] L. Horrein, A. Bouscayrol, Y. Cheng, and C. Dumand, "Hybrid energy management strategy for hybrid electric vehicle," in *Proc. IEEE Vehicle Power Propuls. Conf. (VPPC)*, Oct. 2015, pp. 1–6.